

Group 4 Sellforte: *Detecting Informative Periods in Marketing Data*

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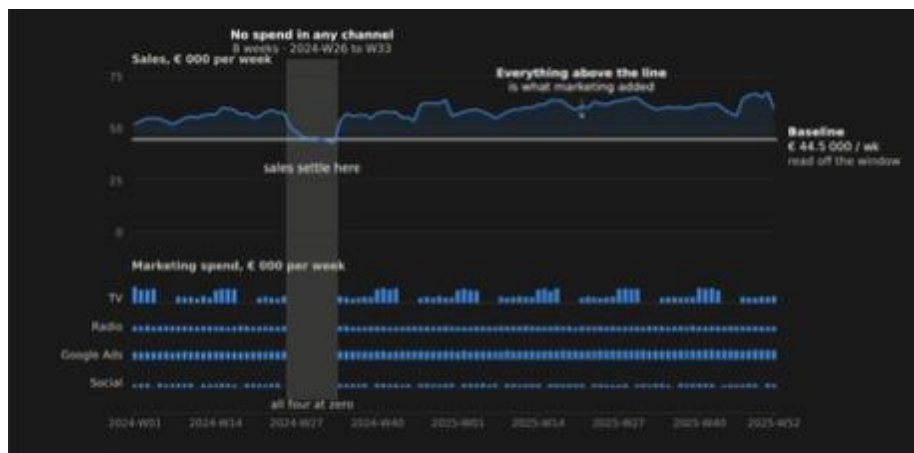
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INTRO:

Marketing measurement usually works with years of data in which everything happens at once: television, radio, search advertising and social media all running together, week after week, while sales move up and down for reasons that include the weather, competitors and the calendar. Untangling this is hard, and much of the difficulty comes from the fact that the marketing channels move together.

But real marketing data is not uniformly difficult. Buried inside it are short periods that are unusually informative — moments where circumstances happened to produce something close to a controlled experiment, without anyone intending it.

The simplest example is the one to start with. If a business had a period with no marketing activity at all, sales during that period tell you the baseline: what the business sells without advertising. Everything else in the model is measured against that number, and here it can simply be read off.



Other patterns are less obvious and more interesting:

- **A single channel goes dark.** One channel stops for six weeks while the others continue. The result behaves like a holdout experiment.
- **A channel pulses on and off.** Advertising has an after-effect: spend stops, but sales stay elevated for a while and then decay. That decay is only visible when spend actually stops, so on–off–on patterns are the only place in the data where the after-effect can be measured.
- **Only one channel is running.** For a few weeks, everything else is off. Whatever moves in sales can be attributed without the usual ambiguity.
- **Spend jumps sharply.** A sudden step up or down in one channel gives a much cleaner read on its effect than a slow drift would.
- **A channel launches at different times in different regions.** Region A gets the campaign in March, region B in June. The regions take turns acting as each other's comparison group.

When advertising is done in multiple regions at once (for example Germany, Austria and Switzerland (DACH)), these patterns may be found either in one, some or all of the markets.

Experienced analysts spot these periods by eye — inconsistently, and only when they think to look. Nobody has systematically catalogued them, and nobody has built a tool that finds them automatically.

That is your project. Build a tool that scans a marketing dataset, finds these informative periods, and presents them to a human being in a way they can inspect and interrogate.

ASSIGNMENT:

Develop algorithms that detect these periods in data. You should be able to detect at least

- Dark period (no marketing activities)
- Single-channel period (only one active marketing channel)
- Natural holdout (one channel drops to zero, others continue)
- Step change (large, sudden changes in marketing budgets)

in one, some or all countries/markets.

Other valuable patterns that you can come up with and detect are also appreciated.

You will receive data samples of media and sales data. Based on these, you should generate synthetic data for testing the algorithms. In the end of the course, Sellforte will provide a full dataset on which the effectiveness of your algorithms will be evaluated on.

To convey your findings to the end user (who may not be familiar with statistics), you must also produce a user interface that can be used to view your findings in an interactive way (ex. Streamlit works for this). Think about ways to rank your findings and emphasize the most important ones. Marketing effectiveness is measured with key performance indicators, such as return on marketing investment, and KPIs such as it can be used to quantify the impact of the marketing channels.

Key deliverables

- Repository with the code to process and analyze the data
- Application to interactively view the findings of your tool
- Technical Documentation: Setup guide for the tool, as well as findings, key learnings and conclusions of this problem field in the final report.

DATA:

Generate appropriate synthetic datasets for yourselves to test your algorithms. Small samples will be provided, so that you can use the correct format and datatypes, matching that of the full datasets used later in evaluation.

Examples of synthetic marketing data generation can be found online, for example

<https://github.com/facebookexperimental/siMMMulator?tab=readme-ov-file> .

TOOLS:

Let's work with Python. You can use the packages you like, for example pandas, numpy, plotly

HINTS:

- The primary goal is come up with clever algorithms
- Test your algorithms on varying datasets, also noisier ones
- In the multi-country case, market size differences can obscure the patterns. Normalizing the data can help in spotting the patterns